

Timing Validation

What Was Tested

The release trigger logic decides when the arm is allowed to let go of the battery, but there's always a small gap between the moment those conditions are confirmed and the moment the release action is carried out. This task was about measuring that gap directly.

I built a standalone ROS2 node in Python that records two timestamps: one for the moment a condition is met, and one for the moment the corresponding action is sent. The node then calculates the time difference between these two points in milliseconds. To test the logging itself, the node was run with a simulated condition-met and action-sent sequence (with a small built-in processing delay) rather than connecting to the live CV pipeline or physical arm.

The node was run using a ROS2 launch/run setup, executing 5 trials automatically, one every 2 seconds.

Results

Across the 5 trials, the measured delay between the condition being met and the action being sent ranged from about 30ms to 45ms, with an average of 40.75ms:

Trial	Delay (ms)
1	30.27
2	43.71
3	43.90
4	40.65
5	45.22
Average	40.75

These results show the timestamp logging and delay calculation work correctly and consistently every trial produced a sensible, similar delay value, with no failures or unexpected jumps.

What This Means

This confirms the timing-measurement tool itself works as intended. It doesn't yet reflect a real trigger condition, the delays measured here come from a simulated processing delay, not an actual velocity or position check.

The next step is to connect this same logging approach to a real trigger condition, so the measured delay reflects genuine system behaviour, and to compare that real delay against the control loop's cycle time to confirm the arm reacts quickly enough in practice.

Evidence

```
jerem@jerem:~/timing_ws$ cd ~/timing_ws
colcon build
Starting >>> timing_validation
Finished <<< timing_validation [2.92s]
```

Summary: 1 package finished [3.45s]

```
jerem@Jerem:~/timing_ws$ ^C
```

```
jerem@Jerem:~/timing_ws$ source install/setup.bash
```

```
jerem@Jerem:~/timing_ws$ ros2 run timing_validation timing_node
```

```
[INFO] [1788947516.503907421] [timing_validation_node]: --- Trial 1 ---
[INFO] [1788947516.505099647] [timing_validation_node]: Condition met at: 1788947516504227791
[INFO] [1788947516.535408017] [timing_validation_node]: Action sent at: 1788947516534498157
[INFO] [1788947516.536541058] [timing_validation_node]: Delay: 30.27 ms
[INFO] [1788947518.454805948] [timing_validation_node]: --- Trial 2 ---
[INFO] [1788947518.455780262] [timing_validation_node]: Condition met at: 1788947518455075148
[INFO] [1788947518.499905870] [timing_validation_node]: Action sent at: 1788947518498788279
[INFO] [1788947518.500764170] [timing_validation_node]: Delay: 43.71 ms
[INFO] [1788947520.455084691] [timing_validation_node]: --- Trial 3 ---
[INFO] [1788947520.455916733] [timing_validation_node]: Condition met at: 1788947520455262548
[INFO] [1788947520.500454390] [timing_validation_node]: Action sent at: 1788947520499164967
[INFO] [1788947520.501435031] [timing_validation_node]: Delay: 43.90 ms
[INFO] [1788947522.457297178] [timing_validation_node]: --- Trial 4 ---
[INFO] [1788947522.458561552] [timing_validation_node]: Condition met at: 1788947522457921034
[INFO] [1788947522.499612497] [timing_validation_node]: Action sent at: 1788947522498573983
[INFO] [1788947522.500644869] [timing_validation_node]: Delay: 40.65 ms
[INFO] [1788947524.455166611] [timing_validation_node]: --- Trial 5 ---
[INFO] [1788947524.456400117] [timing_validation_node]: Condition met at: 1788947524455346609
```

timing_node.py X setup.py package.xml

Ubuntu > home > jerem > timing_ws > src > timing_validation > timing_validation > timing_node.py

```
1  import rclpy
2  from rclpy.node import Node
3  import time
4  import random
5
6  class TimingValidationNode(Node):
7      def __init__(self):
8          super().__init__('timing_validation_node')
9          self.trial_count = 0
10         self.max_trials = 5
11         self.results = []
12
13         # Run one trial every 2 seconds
14         self.timer = self.create_timer(2.0, self.run_trial)
15
16     def run_trial(self):
17         if self.trial_count >= self.max_trials:
18             self.print_results()
19             self.timer.cancel()
20             return
21
22         self.trial_count += 1
23         self.get_logger().info(f'--- Trial {self.trial_count} ---')
24
25         # Step 1: simulate "condition met" (e.g. arm reached pose)
26         t_condition_met = self.get_clock().now()
27         self.get_logger().info(f'Condition met at: {t_condition_met.nanoseconds}')
```

```

1 from setuptools import find_packages, setup
2
3 package_name = 'timing_validation'
4
5 setup(
6     name=package_name,
7     version='0.0.0',
8     packages=find_packages(exclude=['test']),
9     data_files=[
10         ('share/ament_index/resource_index/packages',
11          ['resource/' + package_name]),
12         ('share/' + package_name, ['package.xml']),
13     ],
14     install_requires=['setuptools'],
15     zip_safe=True,
16     maintainer='jerem',
17     maintainer_email='jerem@todo.todo',
18     description='TODO: Package description',
19     license='TODO: License declaration',
20     extras_require={
21         'test': [
22             'pytest',
23         ],
24     },
25     entry_points={
26         'console_scripts': [
27             'timing_node = timing_validation.timing_node:main',
28         ],
29     },

```

```
File Edit Selection View ... Search
timing_node.py x setup.py package.xml x
Ubuntu > home > jerem > timing_ws > src > timing_validation > package.xml
1 <?xml version="1.0"?>
2 <?xml-model href="http://download.ros.org/schema/package_format3.xsd" schematypens="http://www.w3.org/2001/XMLSchema"?>
3 <package format="3">
4   <name>timing_validation</name>
5   <version>0.0.0</version>
6   <description>TODO: Package description</description>
7   <maintainer email="jerem@todo.todo">jerem</maintainer>
8   <license>TODO: License declaration</license>
9
10  <test_depend>ament_copyright</test_depend>
11  <test_depend>ament_flake8</test_depend>
12  <test_depend>ament_pep257</test_depend>
13  <test_depend>python3-pytest</test_depend>
14
15  <export>
16    <build_type>ament_python</build_type>
17  </export>
18  <depend>rcclpy</depend>
19 </package>
20
```